Linyun He

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Education

- **Ph.D. in Operations Research, Minor in Mathematics** *2025*
  Advisor: Dr. Eunhye Song
  Georgia Institute of Technology Atlanta, GA
- **M.A. in Statistics** *2019*
  Columbia University New York, NY
- **B.S. in Mathematics and Applied Mathematics** *2017*
  Fudan University Shanghai, China

Academic Position

- **Postdoctoral Research Fellow** *2025–Present*
  Supervisors: Dr. Carolina Osorio, Dr. Fausto Errico, Dr. Jorge E. Mendoza and Dr. Tommaso Schettini
  Department of Decision Sciences, HEC Montréal Montréal, Canada

Research Domain

- Methodologies: stochastic simulation and optimization, sequential decision making and experimental design, Bayesian optimization, stochastic model validation and calibration, uncertainty quantification
- Applications: digital twins, transportation systems, clinical trials, smart manufacturing

Publications

[1] **Linyun He**, Mingbin Ben Feng, and Eunhye Song (2026). Efficient Input Uncertainty Quantification for Ratio Estimator. *INFORMS Journal on Computing*. Articles in Advance.

[2] **Linyun He**, Luke Rhodes-Leader, and Eunhye Song (2026). Validating a Stochastic Digital Twin with Multi-Epoch, Multivariate Data. *INFORMS Journal on Data Science*. Forthcoming.

[3] **Linyun He**, Uday V. Shanbhag, and Eunhye Song (2024). Stochastic Approximation for Multi-period Simulation Optimization with Streaming Input Data. *ACM Transactions on Modeling and Computer Simulation*, 34(2):1–27.

[4] Zhunxuan Wang, **Linyun He**, Chunchuan Lyu, and Shay B. Cohen (2022). Nonparametric Learning of Two-Layer ReLU Residual Units. *Transactions on Machine Learning Research*, 1–41.

[5] Zihao Wang, **Linyun He**, Zhenyun Qin, Roger Grimshaw, and Gui Mu (2019). High-order Rogue Waves and Their Dynamics of the Fokas–Lenells Equation Revisited: A Variable Separation Technique. *Nonlinear Dynamics*, 98:2067–2077.

[6] **Linyun He** and Eunhye Song (2024). Introductory Tutorial: Simulation Optimization under Input Uncertainty. In *Proceedings of the 2024 Winter Simulation Conference (WSC)*, 1338–1352.

[7] **Linyun He**, Luke Rhodes-Leader, and Eunhye Song (2024). Digital Twin Validation with Multi-Epoch, Multi-Variate Output Data. In *Proceedings of the 2024 Winter Simulation Conference (WSC)*, 347–358.

[8] **Linyun He**, Eunhye Song, and Ben Feng (2023). Efficient Input Uncertainty Quantification for Regenerative Simulation. In *Proceedings of the 2023 Winter Simulation Conference (WSC)*, 385–396.

[9] **Linyun He** and Eunhye Song (2021). Nonparametric Kullback-Liebler Divergence Estimation Using M-Spacing. In *Proceedings of the 2021 Winter Simulation Conference (WSC)*, 1–12.

Preprints and Working Papers

[1] **Linyun He**, Fausto Errico, Jorge E. Mendoza, Carolina Osorio, and Tommaso Schettini (2026). Combinatorial Bayesian Optimization with Low-Cost Approximation Functions. In preparation.

[2] **Linyun He**, Ozge Yapar, and Yuwei Zhou (2026). Adaptive Dose Optimization Across Multiple Populations in Basket Trials. In preparation.

Awards and Honors

[1] Outstanding Reviewer Award of the 2025 Winter Simulation Conference Proceedings 2025
[2] Winner of the 18th GERAD Postdoctoral Fellowship Competition 2025
[3] Finalist of the 2023 Winter Simulation Conference Best Theoretical Contributed Paper Award **(5/209)** 2023
[4] Overseas Exchange Scholarship by Fudan University Education Development Foundation 2017
[5] Third Prize of the Scholarship for Outstanding Students at Fudan University 2014, 2015, 2016
[6] Major Scholarship of School of Mathematical Sciences at Fudan University 2014, 2015, 2016

Presentations

[1] GERAD Seminar (Invited) Montréal, Canada, *2026.09*
Title: Combinatorial Bayesian Optimization with Low-Cost Approximation Functions
[2] Optimization Days (JOPT) 2026 Montréal, Canada, *2026.05*
Title: Combinatorial Bayesian Optimization with Low-Cost Approximation Functions
[3] 2025 INFORMS Annual Meeting Atlanta, GA, *2025.10*
Title: Simulation Digital Twin Validation with Multi-Epoch Key Performance Indicators
[4] 2024 Winter Simulation Conference Orlando, FL, *2024.12*
Title a: Digital Twin Validation with Multi-Epoch, Multi-Variate Output Data
Title b: Introductory Tutorial: Simulation Optimization under Input Uncertainty
[5] 2024 INFORMS Annual Meeting Seattle, WA, *2024.10*
Title: Digital Twin Validation with Multi-Epoch, Multi-Dimensional Data
[6] 2023 Winter Simulation Conference San Antonio, TX, *2023.12*
Title: Efficient Input Uncertainty Quantification for Regenerative Simulation
[7] 2023 INFORMS Annual Meeting Phoenix, AZ, *2023.10*
Title: Efficient Input Uncertainty Quantification for Regenerative Simulation
[8] 2023 IISE Annual Conference New Orleans, LA, *2023.05*
Title: Efficient Input Uncertainty Quantification for Steady-state Simulation
[9] 2022 IISE Annual Conference Seattle, WA, *2022.05*
Title: Statistical Analysis of Data-driven Simulation: Model Calibration and Optimization
[10] 2021 Winter Simulation Conference Phoenix, AZ, *2021.12*
Title: Nonparametric Kullback-Liebler Divergence Estimation Using M-Spacing
[11] 2020 INFORMS Annual Meeting online, *2020.11*
Title: Stochastic Approximation for Simulation Optimization with Streaming Input Data

Teaching Experience

- *Instructor of Record* for ISYE3044 Simulation Analysis and Design Georgia Tech, summer 2023
- *Teaching Assistant* for IE425 Stochastic Models in Operations Research Penn State University, spring 2021
- *Teaching Assistant* for IE453 Simulation Modeling For Decision Support Penn State University, fall 2020

Industrial Experience

- *Research Scientist Intern* at **Amazon.com, Inc.** Bellevue, WA *2022.05–2022.08*
- *Quantitative Analyst Intern* at **COFCO Futures Co., LTD** Shanghai, China *2017.01–2017.03*
- *Quantitative Analyst Intern* at **Everbright Securities Co., LTD** Shanghai, China *2016.06–2016.08*

Service

- Reviewer for Mathematics of Operations Research, INFORMS Journal on Computing and Winter Simulation Conference
- Session Chair for 2024 INFORMS Annual Meeting and 2023–2024 Winter Simulation Conferences